Data-Driven Insights

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- Methodology
- Results
- Conclusion
- Appendix/Additional Insights

Executive Summary

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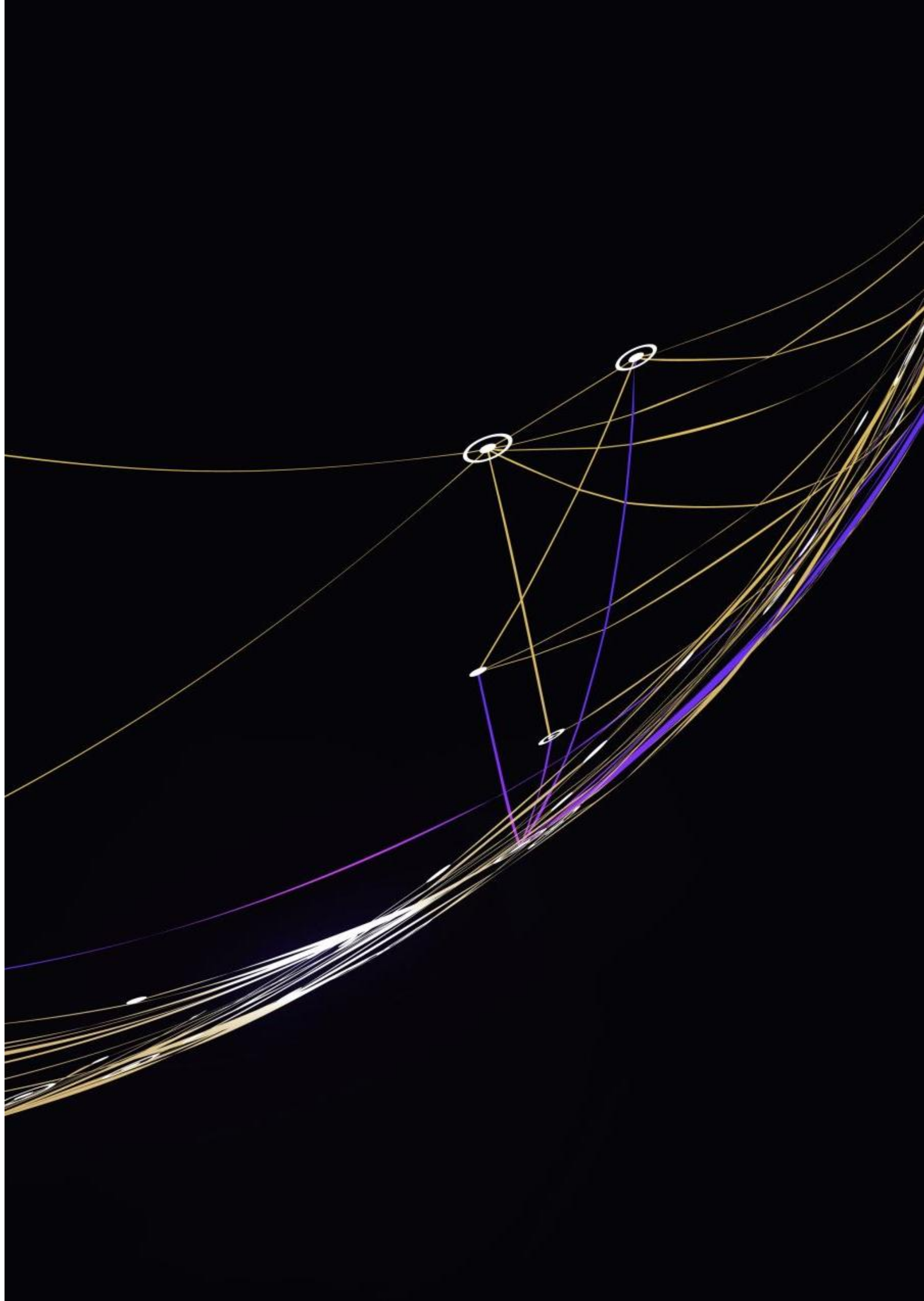
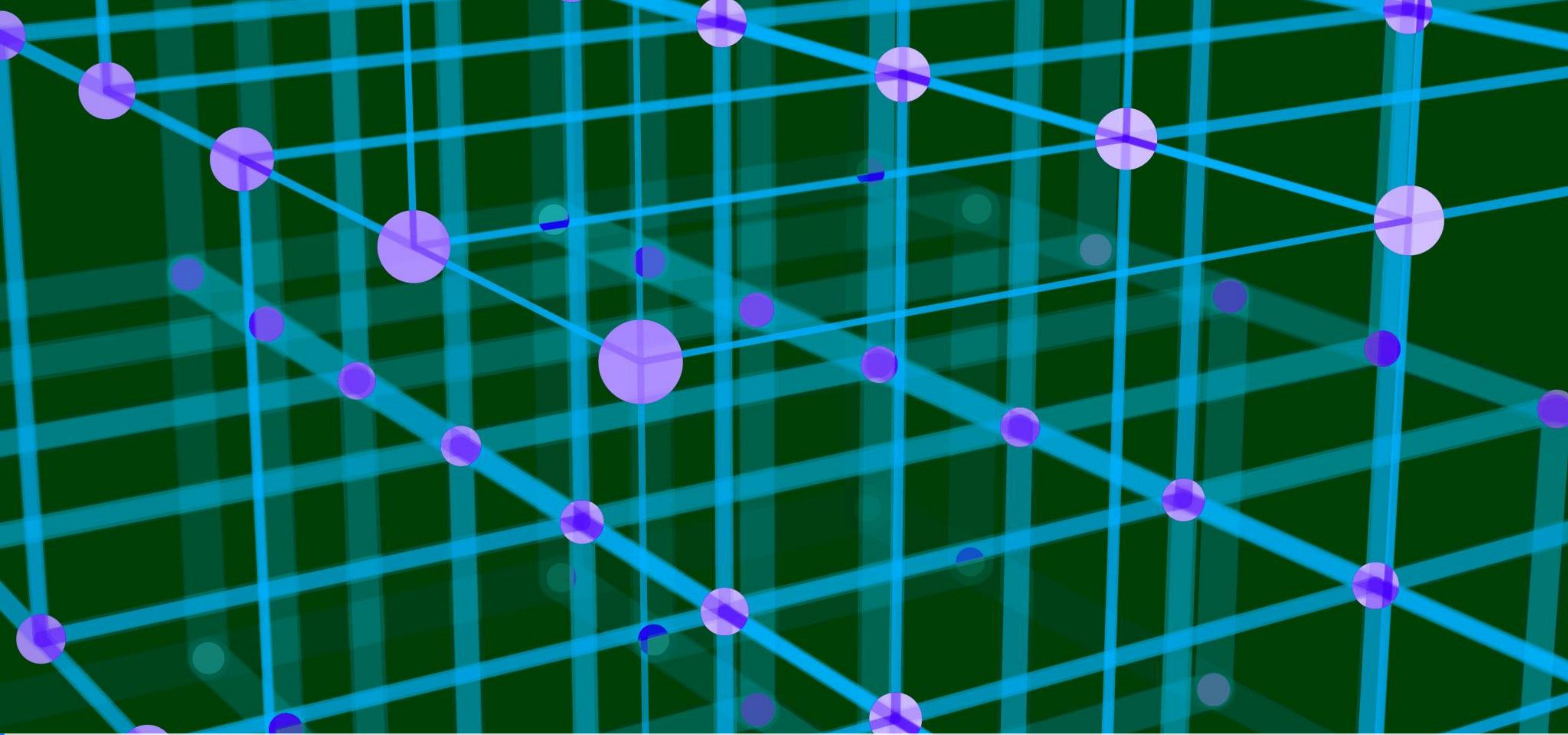
  - EDA With Data visualization
  - EDA With SQL
  - Building an interactive map
  - Building an interactiv dashboard
  - Predictive análisis
  - Conclusion
- Summary of all results
  - EDA Identified Best features to predict
  - Machine Learning Model

Introduction

- Project background and context In this project, the goal is to predict if the Falcon 9 first stage will land successfully. The context revolves around predicting the success of Falcon 9 first stage landings, which is crucial in determining the cost of a launch. SpaceX, offering launches at a significantly lower cost compared to other providers, achieves these savings through the reuse of the first stage. Consequently, understanding the determinants of successful landings can be valuable for companies competing with SpaceX in bidding for rocket launches.





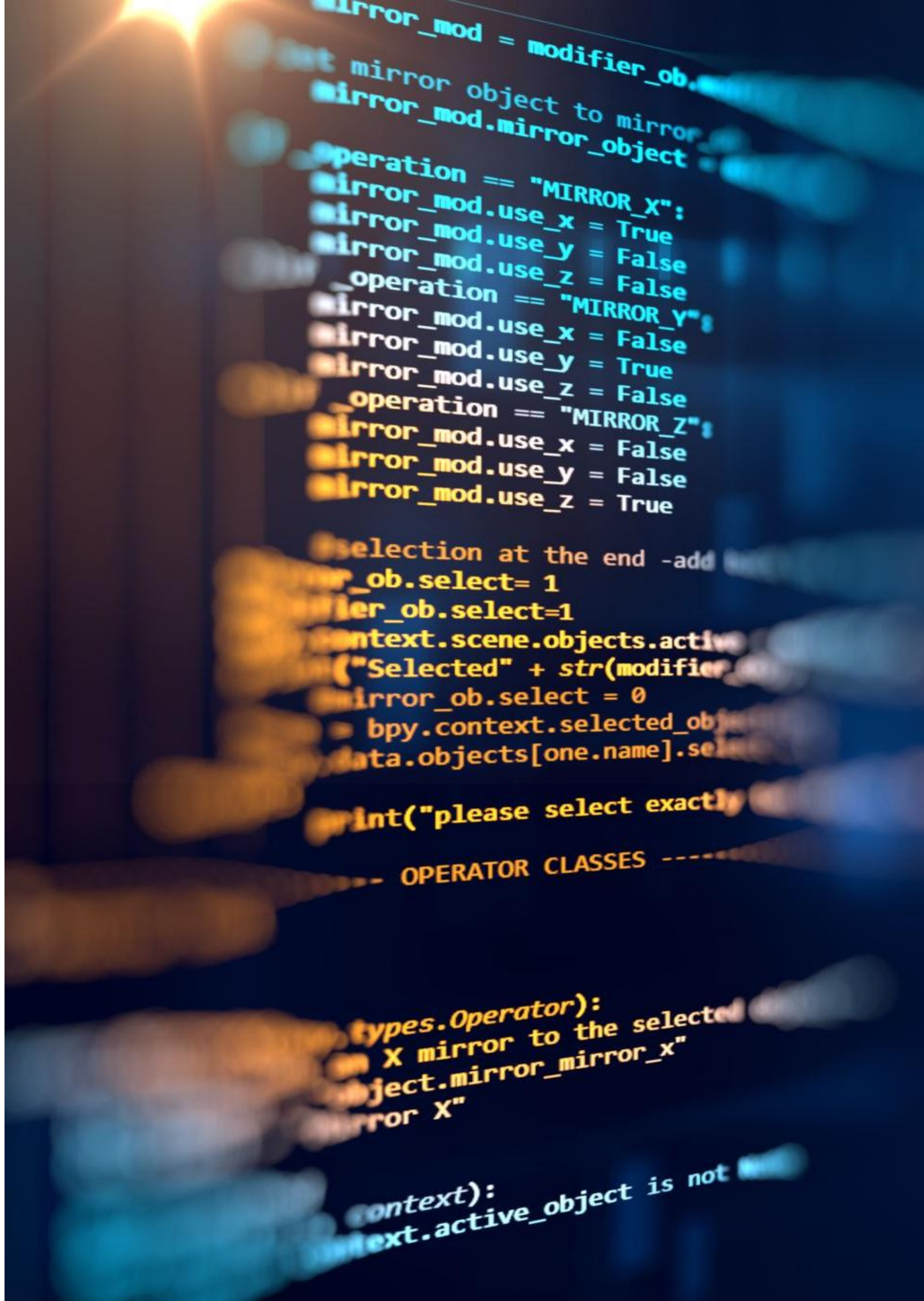
Methodology

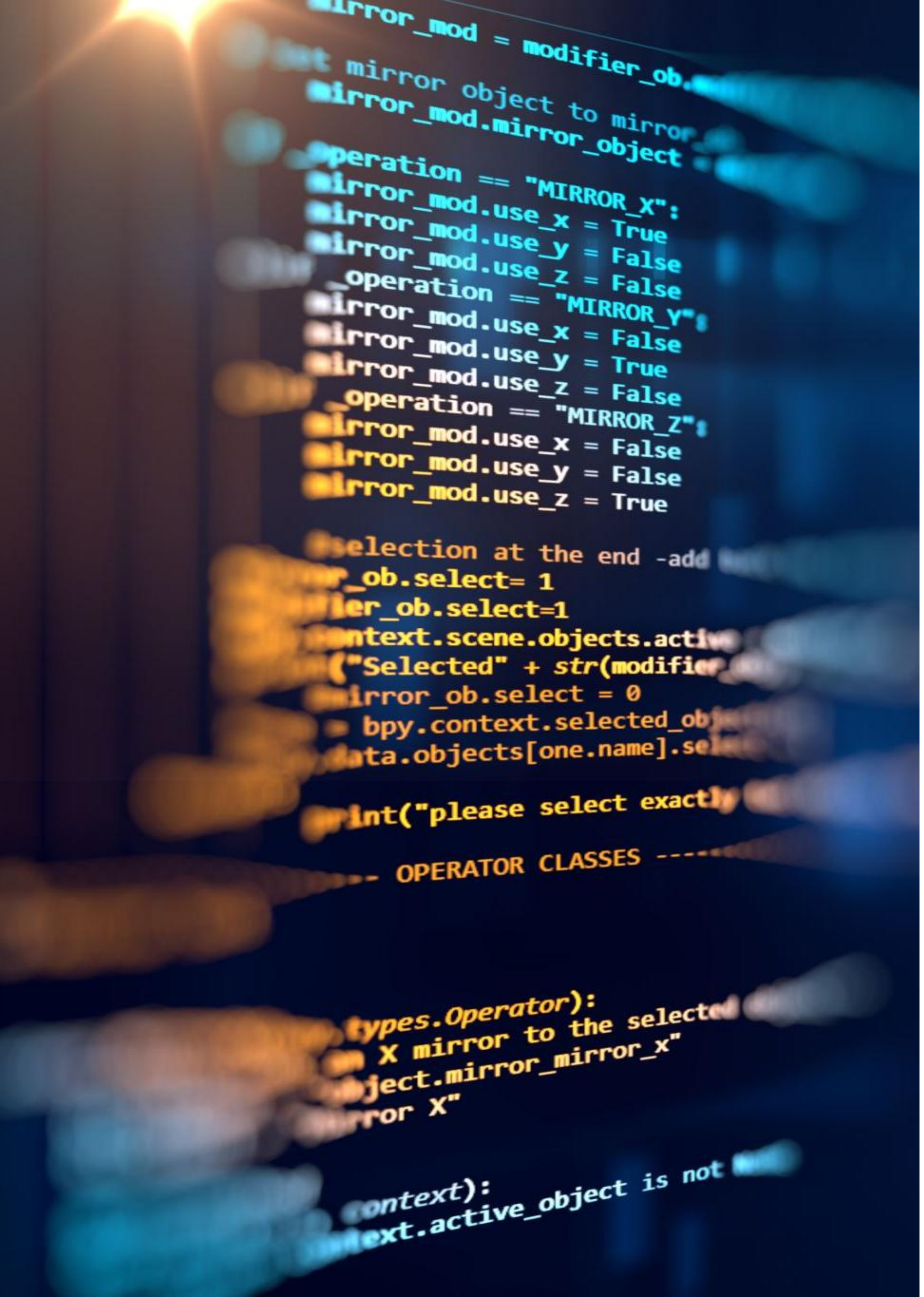
SECTION 1

Data collection

- We collected data sets from SpaceX API and, via web scraping techniques from the Falcon9 pages given by IBM.

- To gather launch-related information, we utilized the API once again by querying the launch IDs. Specifically, we focused on extracting data from the following columns: rocket, payloads, launchpad, and cores.





Data Prepared and transformed

When we collected the raw SpaceX data (from API, CSV, SQL), it was **not ready for machine learning or analysis**. The preparation process involved several steps:

1. Handling Missing Values
2. Standardizing Formats
3. Encoding Categorical Variables
4. Feature Engineering
5. Scaling Numerical Features
6. Final Transformed Dataset

EDA With Data visualization

Scatter Charts:

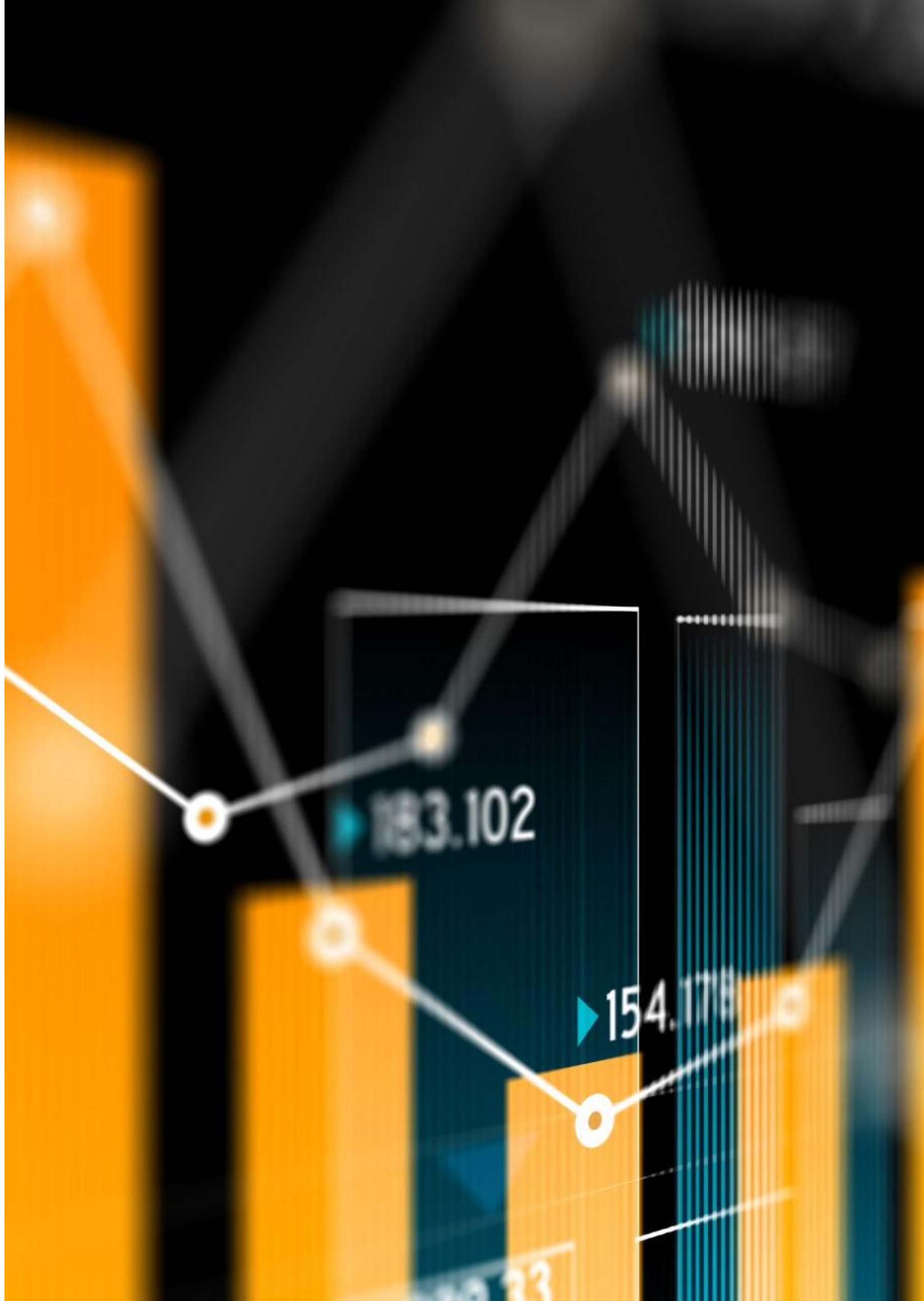
1. Flight Number vs. Launch Site: Identify successful launch sites and observe trends over time.
2. Payload vs. Launch Site: Determine launch site requirements based on payload weight. Flight Number vs. Orbit Type: Analyze mission outcome by orbit type over time.
3. Payload vs. Orbit Type: Investigate if payload weight influences optimal orbit.

Bar Chart:

Orbit vs Success Rate: Determine which orbits have higher success rates.

Line Chart:

Launch Success Trend: Track the yearly trend of average success rates.



EDA with SQL

- Date of first successful landing on ground pad
- Names of boosters
- Total number of successful and failed missions
- Names of booster versions which have carried the max payload
- Failed landing outcomes on drone ship
- Count of landing Outcome

Build an interactive Map

- Markers indicating Launch Sites
- Map with folium
- Markers of launch Outcomes
- Distances Between LaunchSite and proximities.

Build an interactive Dashboard

- Launch Site Dropdown: Enables users to select either all launch sites or a specific launch site for analysis.
- Payload Mass Range Slider: Allows users to define the range of payload masses to consider in the analysis.
- Pie Chart: Success Rates. Provides a visual representation of successful and unsuccessful launches as a percentage of the total, based on the selected launch site or all sites.
- Scatter Chart: Payload Mass vs. Success Rate by Booster Version: presents a scatter chart that demonstrates the correlation between payload mass and launch success, categorized by booster version.

Predictive Analysis

Utilize GridSearchCV for parameter optimization, with a cross-validation value (cv) of 10.

Apply GridSearchCV on various algorithms: Logistic Regression

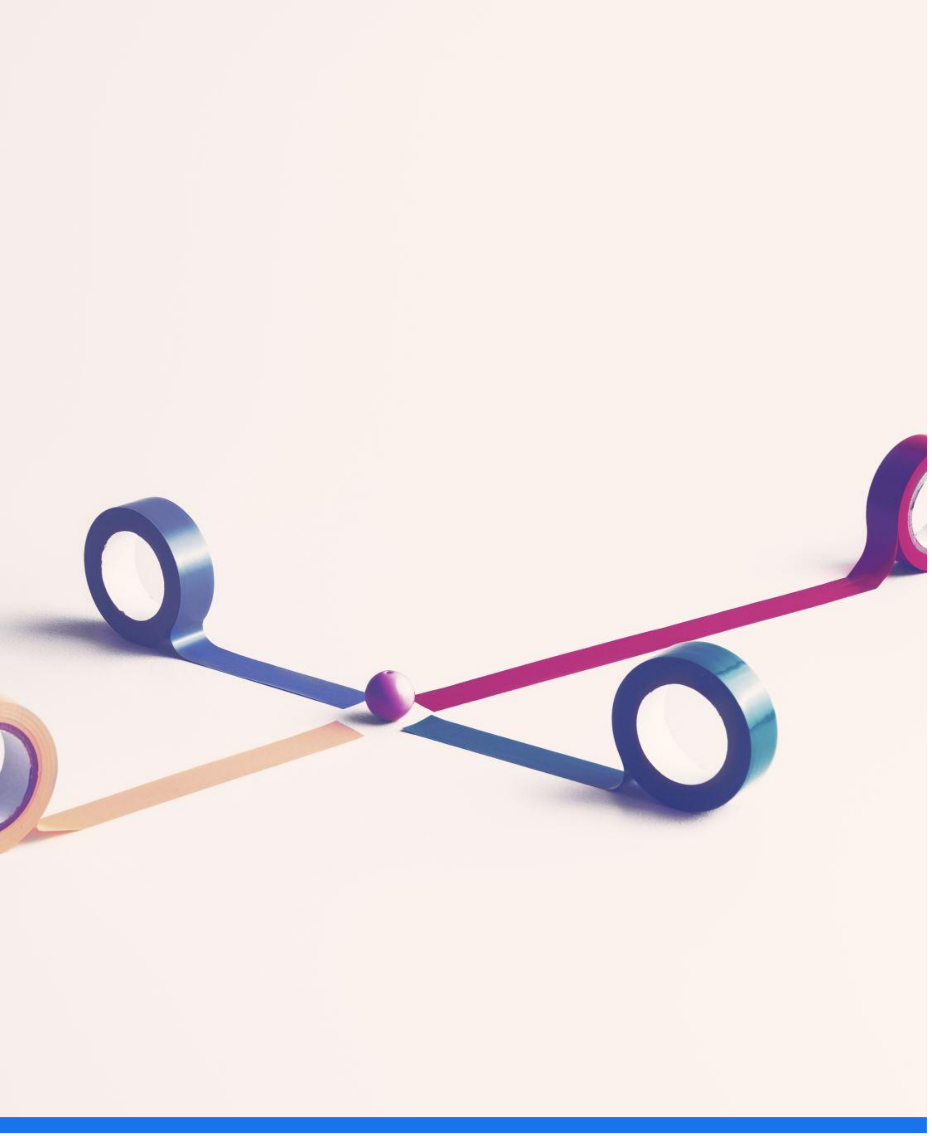
(LogisticRegression()), Support Vector Machine (SVC()), Decision Tree

(DecisionTreeClassifier()), and K-Nearest Neighbors (KNeighborsClassifier()).

Calculate the accuracy of all models on the test data using the .score() function.

Assess the confusion matrix for each model.

Determine the best model based on evaluation metrics: Jaccard Score, F1 Score, and Accuracy.



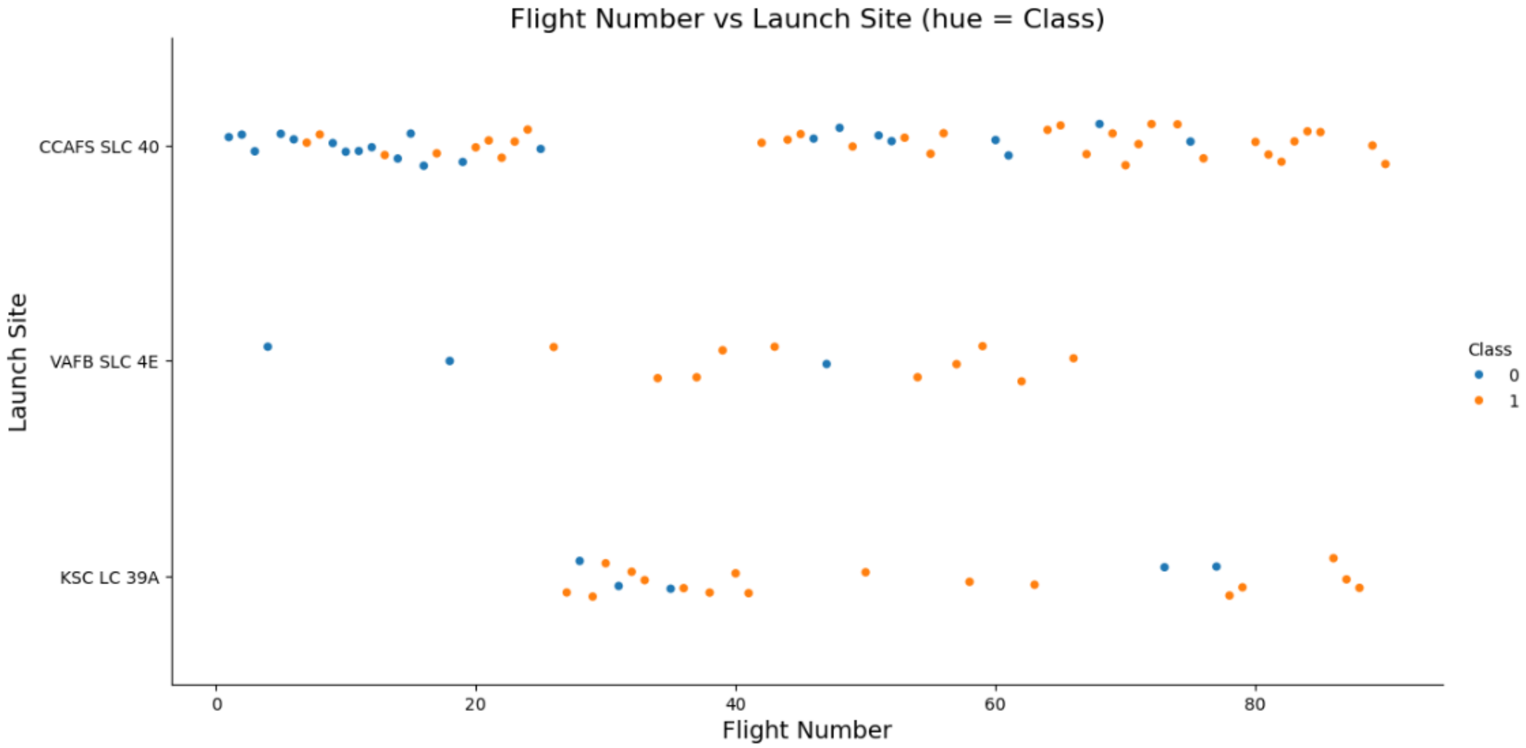
Conclusions

Model Performance: the models exhibited similar performance on the test set, with the decision tree model slightly outperforming the others. It can be employed to predict successful landings, ultimately contributing to increased profits.

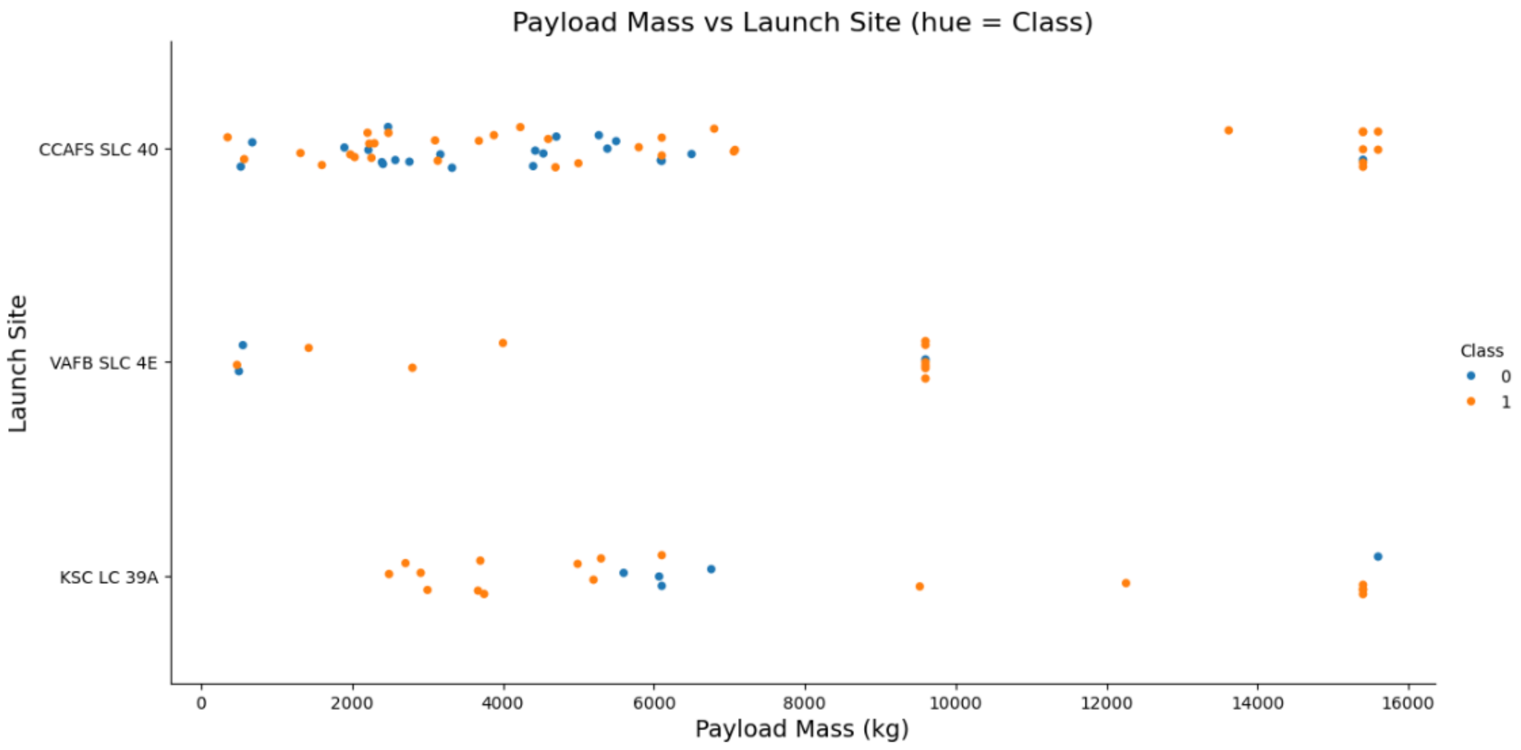
Key Findings:

- Equator Proximity: Most launch sites are strategically located near the equator, leveraging the rotational speed of the Earth to save on fuel and boosters.
- Coastal Proximity: All launch sites are situated in close proximity to coastlines.
- Launch Success: The overall launch success rate has shown a positive trend over time.
- KSC LC-39A: This launch site demonstrated the highest success rate among all sites, particularly for launches with payloads weighing less than 5,500 kg.
- Orbit Success: Orbits including ES-L1, GEO, HEO, and SSO achieved a 100% success rate.
- Payload Mass: Across all launch sites, there is a positive correlation between higher payload mass (kg) and higher success rates.

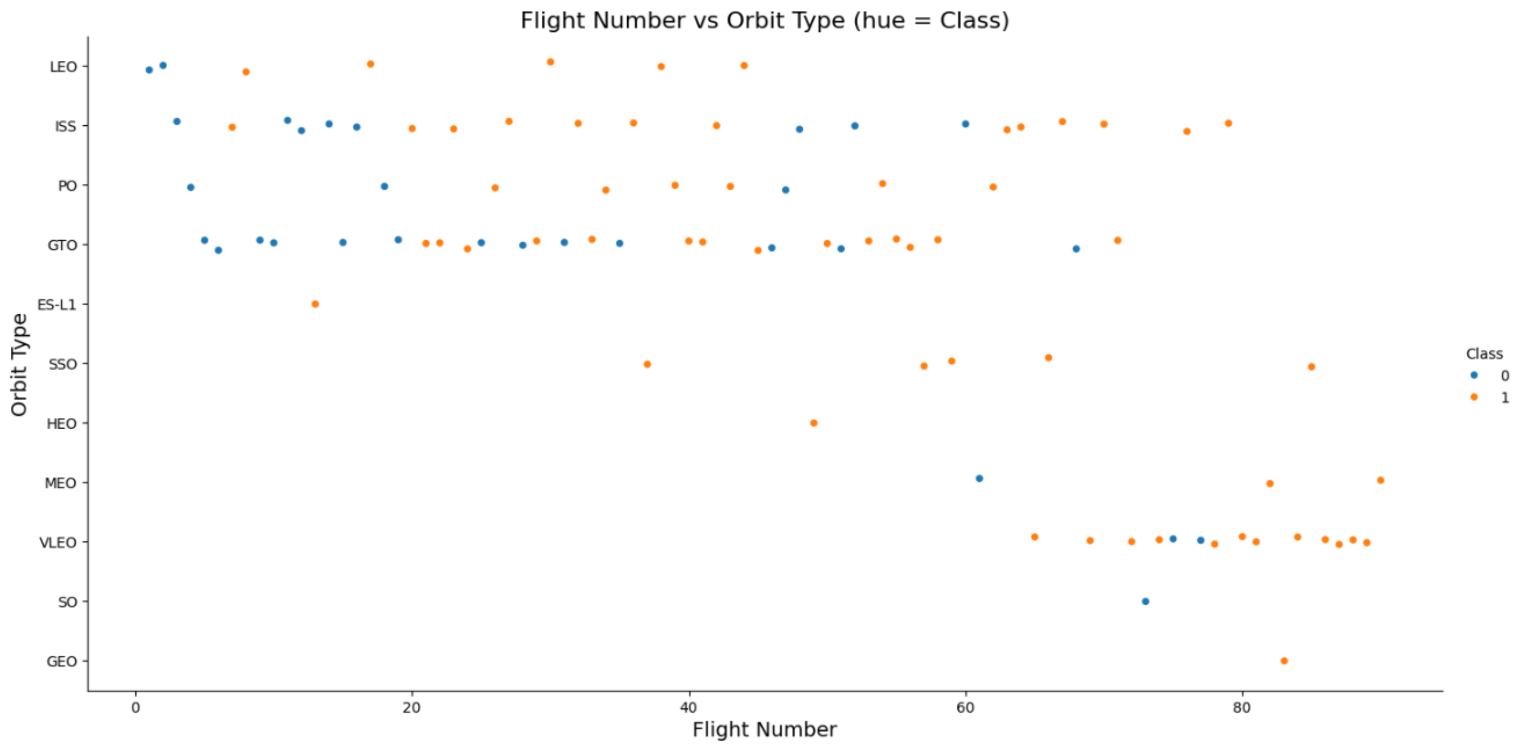
Plots: Flight Number vs. Launch Site



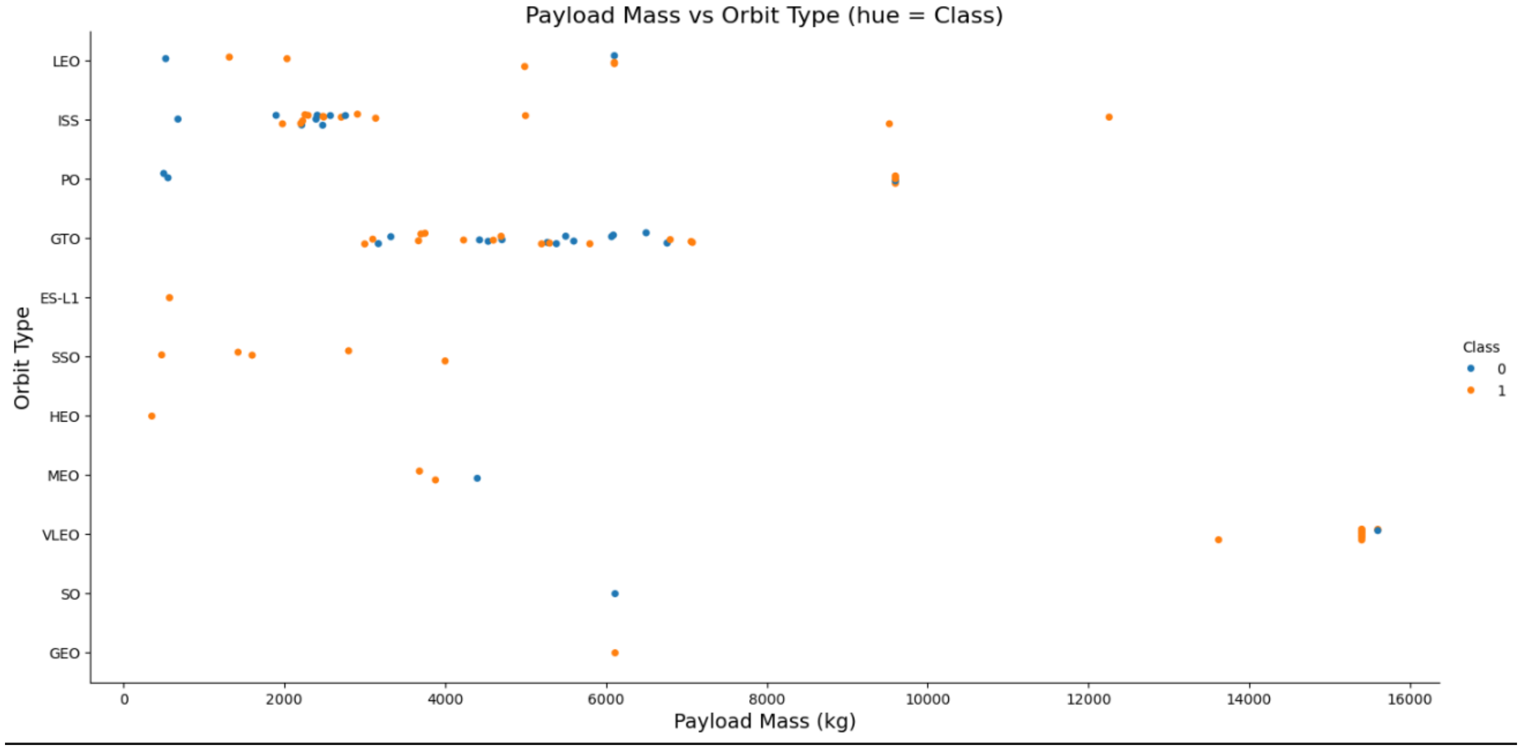
Plots:Payloadmass vs. Launch Site



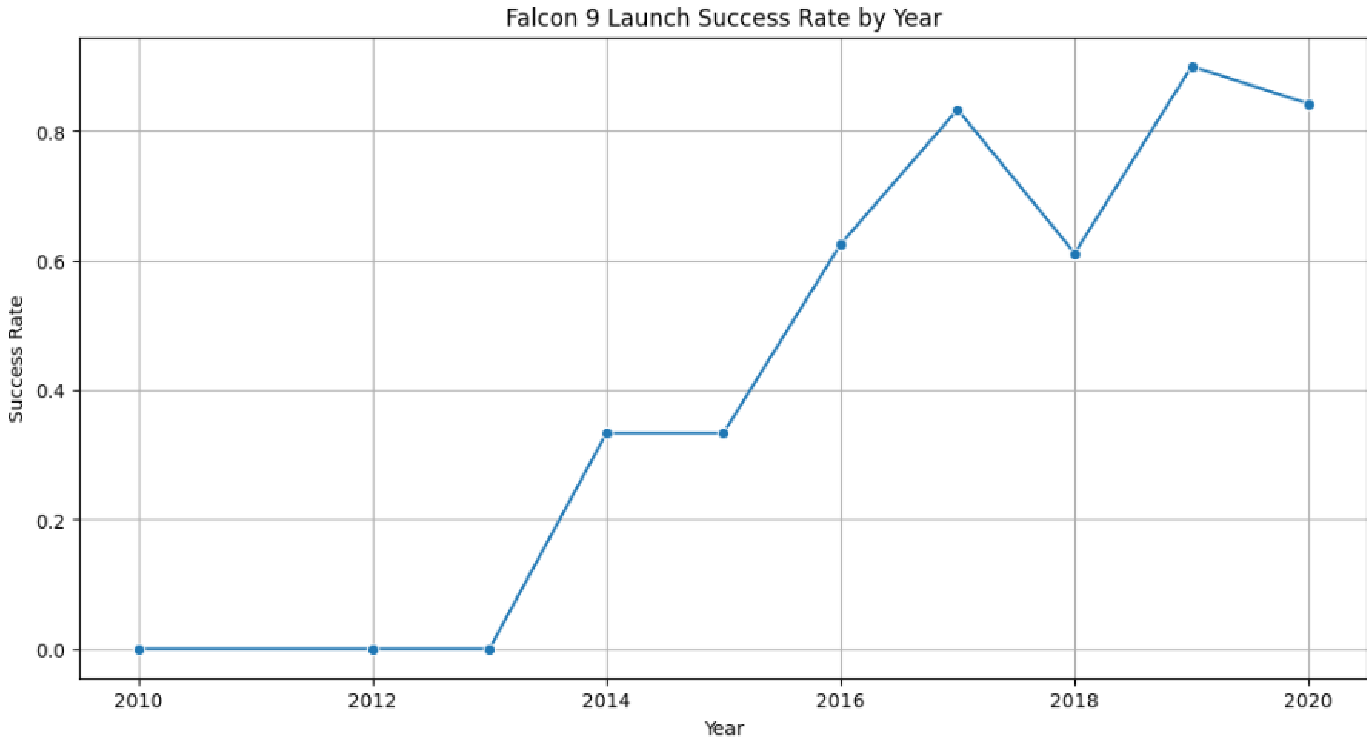
Plots: Flight Number vs. Orbit Type



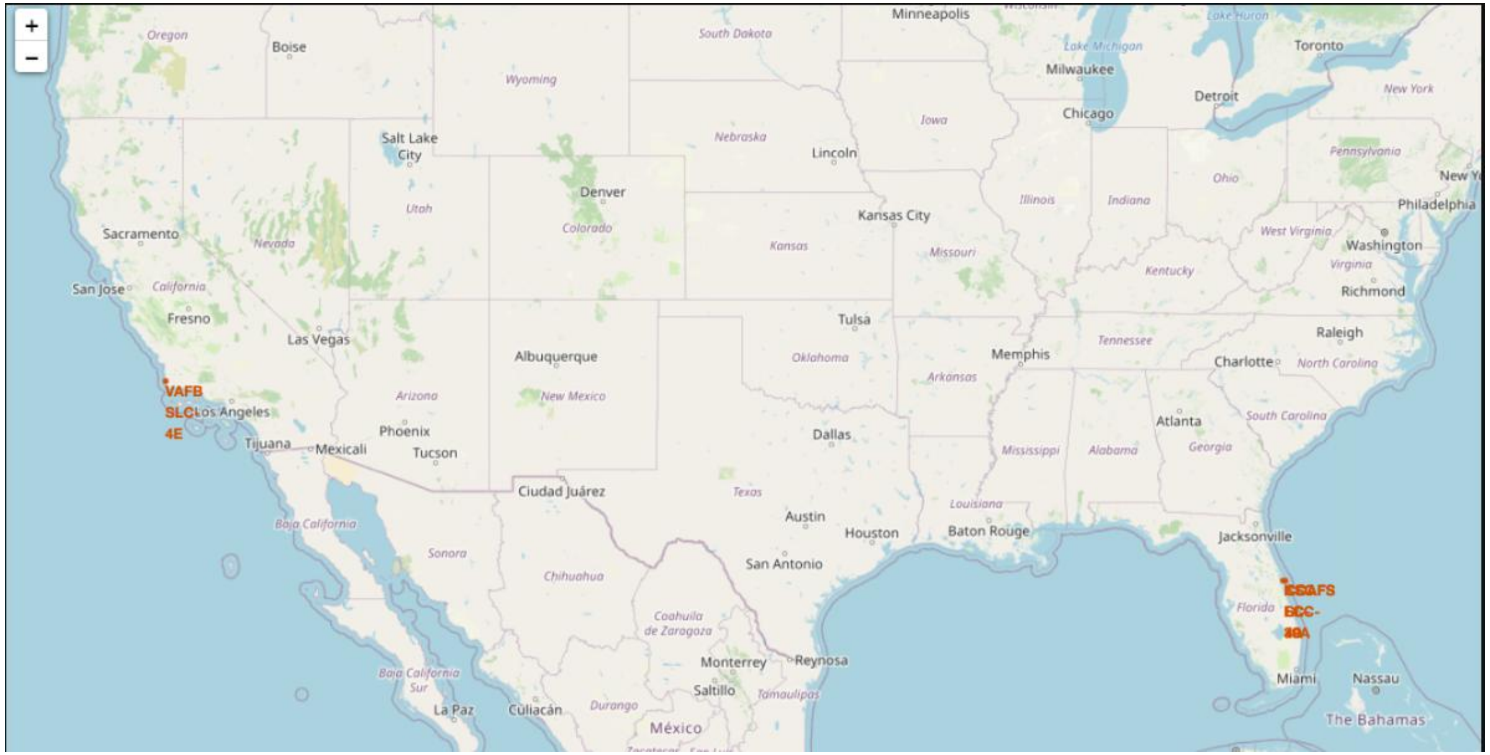
Plots: Payload vs. Orbit Type



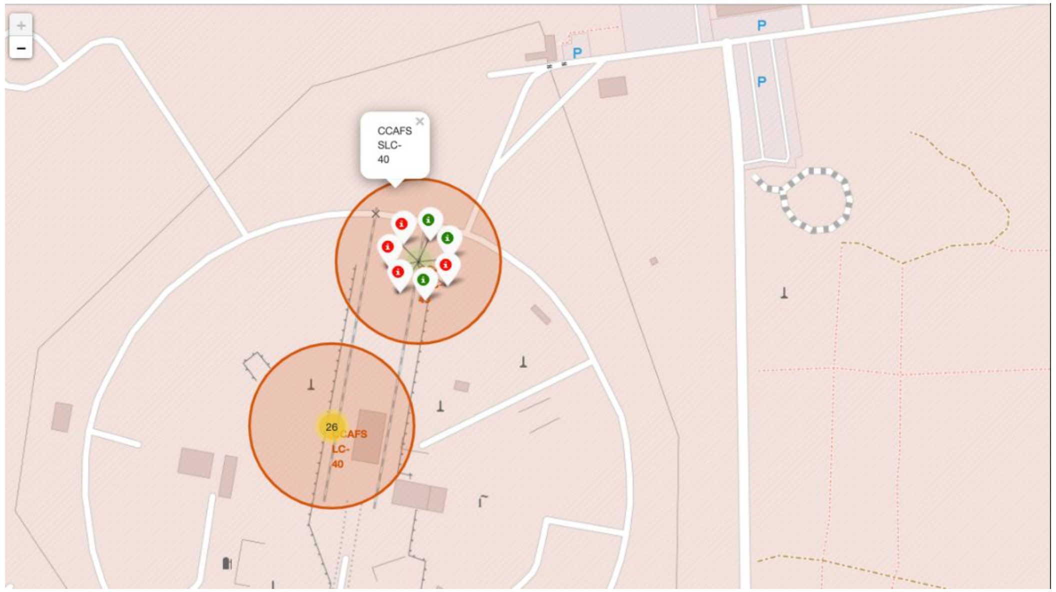
Plots: Launch Success Yearly Trend

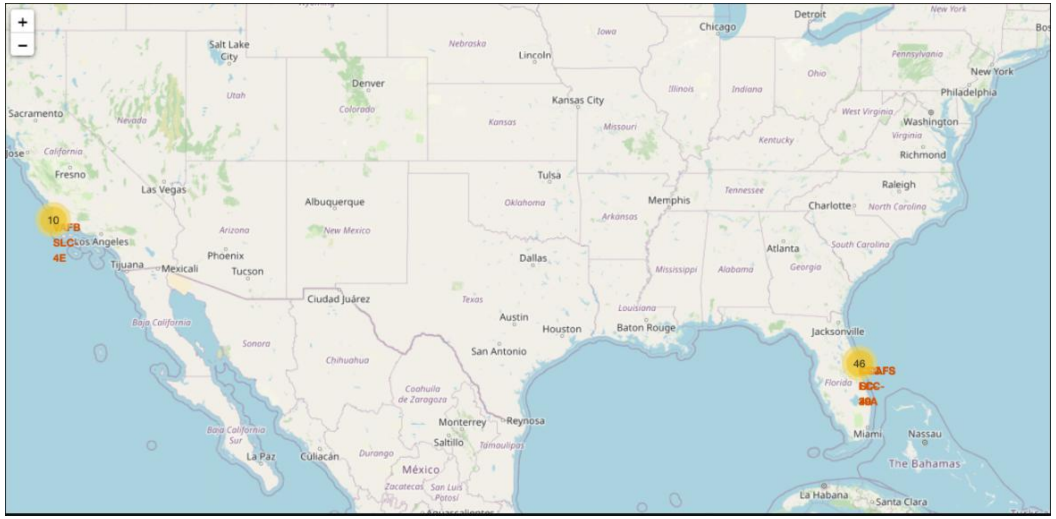


Interactive Map



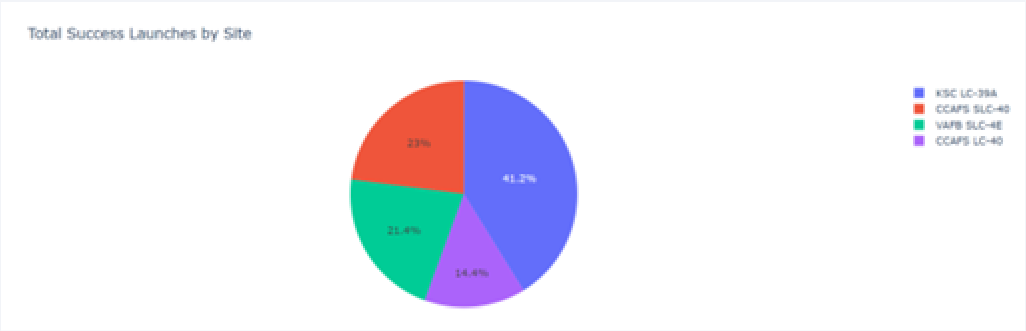
Interactive Map



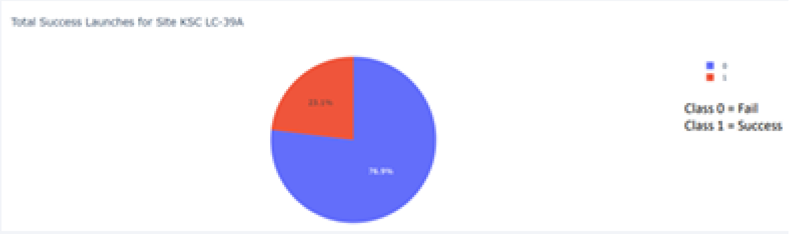


Calculating distance

Build a Dashboard



KSC LC-39A is the most successful launch site (41.2%) with the highest success rate score.



Build a dashboard

Payloads between 2,000 kg and 5,000 kg have the highest success rate

Classification Accuracy

```
jaccard_scores = [
                  jaccard_score(Y, logreg_cv.predict(X), average='binary'),
                  jaccard_score(Y, svm_cv.predict(X), average='binary'),
                  jaccard_score(Y, tree_cv.predict(X), average='binary'),
                  jaccard_score(Y, knn_cv.predict(X), average='binary'),
                 ]

f1_scores = [
             f1_score(Y, logreg_cv.predict(X), average='binary'),
             f1_score(Y, svm_cv.predict(X), average='binary'),
             f1_score(Y, tree_cv.predict(X), average='binary'),
             f1_score(Y, knn_cv.predict(X), average='binary'),
            ]

accuracy = [logreg_cv.score(X, Y), svm_cv.score(X, Y), tree_cv.score(X, Y), knn_cv.score(X, Y)]

scores = pd.DataFrame(np.array([jaccard_scores, f1_scores, accuracy]),
                      index=['Jaccard_Score', 'F1_Score', 'Accuracy'],
                      columns=['LogReg', 'SVM', 'Tree', 'KNN'])

scores
```

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.833333	0.845070	0.805556	0.819444
F1_Score	0.909091	0.916031	0.892308	0.900763
Accuracy	0.866667	0.877778	0.844444	0.855556

	Logistic Regression	SVM	Decision Tree	KNN
Jaccard Score	0.800000	0.800000	0.800000	0.800000
F1 Score	0.888889	0.888889	0.888889	0.888889
Accuracy	0.833333	0.833333	0.833333	0.833333


```
jaccard_scores = [
                  jaccard_score(Y, logreg_cv.predict(X), average='binary'),
                  jaccard_score(Y, svm_cv.predict(X), average='binary'),
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             f1_score(Y, svm_cv.predict(X), average='binary'),
             f1_score(Y, tree_cv.predict(X), average='binary'),
             f1_score(Y, knn_cv.predict(X), average='binary'),
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                      index=['Jaccard_Score', 'F1_Score', 'Accuracy'],
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scores
```
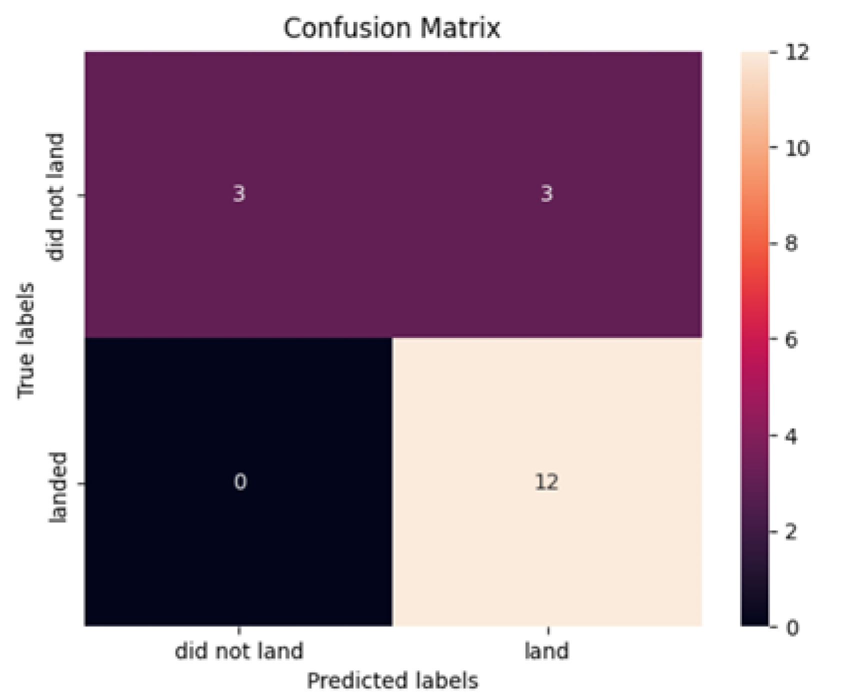


- The accuracy is extremely close, this is likely due to the small dataset.

Confusion Matrix



All the confusion matrices were identical

There are 3 false positive, which is not a good news.

Additional Insights

- Lower weighted payloads tend to exhibit better success rates compared to heavier payloads.
- The success rates of SpaceX launches increase proportionally with the passage of time, indicating a progressive improvement in launch performance.
- The Decision Tree Classifier can be employed to predict successful landings, ultimately contributing to increased profits.
- A larger dataset will help build on the predictive analytics results to help understand if the findings can be generalizable to a larger data set.